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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
63/965,450	01/22/2026		

CONFIRMATION NO. 7115

IMPROPER CFR REQUEST

OC000000102033651

221639
Eric Haywood
149 Remington Street Apt 425
Fort Collins, CO 80524

Date Mailed: 02/26/2026

RESPONSE TO REQUEST FOR CORRECTED FILING RECEIPT

Power of Attorney, Claims, Fees, System Limitations, and Miscellaneous

In response to your request for a corrected filing Receipt dated 01/22/2026, the Office is unable to comply with your request because:

- The ADS received on 01/22/2026 was not properly signed. Therefore, the Office will treat it only as a transmittal letter. See 37 CFR 1.76(e). Inventorship has not been set by this document and any foreign priority or domestic benefit claims contained therein are ineffective. See 37 CFR 1.55 or 37 CFR 1.78.

  If the applicant wishes to submit another ADS:
- It must be properly signed by a party under 37 CFR 1.33(b), and be signed in compliance with 37 CFR 1.4(d). Note that all papers submitted on behalf of a juristic entity must be signed by a patent practitioner unless otherwise specified.
- Changes to the information of record must be properly marked up in compliance with 37 CFR 1.76(c), i.e. must identify the information that is being changed, with underlining for insertions, and strike-through or brackets for text removed. In general, the identification of the information being changed should be made relative to the most recent filing receipt.
- Benefit and priority claims must be presented in an ADS in compliance with 37 CFR 1.76(c) and within the time periods specified in 37 CFR 1.55 and 1.78.

  If an ADS as set forth above is provided, the filing of the inventor's oath or declaration may be postponed until the application is otherwise in condition for allowance. See 37 CFR 1.53(f). Note that the inventor's oath or declaration must be filed no later than the date on which the issue fee has been paid.

How to Make Changes in the Information of Record Using a Corrected Application Data Sheet: In order to make changes to the information of record, an ADS must be properly signed and properly marked up relative to the current information of record.

Proper signature: The ADS must be signed with a handwritten signature or proper S-signature by:

- A patent practitioner, with the practitioner's registration number accompanying the signature (e.g., immediately below or adjacent to the signature), or
- The applicant, if the applicant is an individual other than the inventor(s) and no power of attorney has been appointed, or
- All of the inventors, if no other applicant has been established and no power of attorney has been appointed.

A proper S-signature consists of only letters and/or Arabic numerals, with appropriate spaces and commas, periods, apostrophes, or hyphens for punctuation contained between a first single forward slash mark before, and a second single forward slash mark after, the S-signature.

Proper markings: The ADS must identify the changes being made with underlining for insertions and strike-through or brackets for text removed relative to the information of record. No other markings or indications are acceptable. Where an ADS providing corrected or updated information does not contain all of the sections of the ADS, the entire section (including the heading) in which changes are being made must be included in the ADS. Information of record can generally be found on the latest filing receipt.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/slee/

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY.DOCKET.NO	TOT CLAIMS	IND CLAIMS
63/965,450	01/22/2026		0			

CONFIRMATION NO. 7115

FILING RECEIPT

OC000000102033560

221639
Eric Haywood
149 Remington Street Apt 425
Fort Collins, CO 80524

Date Mailed: 02/26/2026

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt identifying the requested changes, preferably by including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

None

Applicant(s)

Power of Attorney: None

Permission to Access Application via Priority Document Exchange: No

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 02/24/2026

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 63/965,450**

Projected Publication Date: None

Non-Publication Request: No

Early Publication Request: No

Title

System Efficiency Optimization via Spectral Analysis of Operational Frequencies and Constructive Interference Tuning

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at http://www.uspto.gov/web/offices/pac/doc/general/index.html.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, http://www.stopfakes.gov. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign AssetsControl, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit http://www.SelectUSA.gov or call +1-202-482-6800.

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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
63/965,450	01/22/2026		

CONFIRMATION NO. 7115

221639
Eric Haywood
149 Remington Street Apt 425
Fort Collins, CO 80524

OC000000102033562

Date Mailed: 02/26/2026

Improper Submission of Authorization to Permit Access to Application by Participating Offices under 37 CFR 1.14(h)(1) (Priority Document Exchange)

The Authorization to Permit Access to Application-As-Filed by Participating Offices under 37 CFR 1.14(h)(1) (authorization to permit access to application via priority document exchange) submitted on 01/22/2026 in the above-identified application is not accepted because:

- It was not properly signed. If applicant still wishes to provide authorization to permit access under 37 CFR 1.14(h)(1), applicant must submit a properly signed authorization (e.g., PTO/SB/39).

Any authorization should be submitted prior to filing a subsequent foreign application with a participating intellectual property office in which priority is claimed to the above-identified U.S. application to ensure that it is likely that the participating foreign intellectual property office will be successful in its attempt to retrieve a copy of the U.S. priority application from the Office.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/slee/

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
63/965,450	01/22/2026		

CONFIRMATION NO. 7115

221639
Eric Haywood
149 Remington Street Apt 425
Fort Collins, CO 80524

OC000000102033564

Date Mailed: 02/26/2026

Improper Submission of Authorization to Permit Access to Search Results under 37 CFR 1.14(h)(2)

The Authorization to Permit Access to the Search Results from the instant application under 37 CFR 1.14(h)(2) (authorization to permit access to search results) submitted on 01/22/2026 in the above-identified application is not accepted because:

- It was not properly signed. If applicant still wishes to provide authorization to permit access to the search results under 37 CFR 1.14(h)(2), applicant must submit a properly signed authorization (e.g., PTO/SB/69).

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/slee/

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
63/965,450	01/22/2026		

CONFIRMATION NO. 7115

FORMALITIES LETTER

OC000000102033561

221639
Eric Haywood
149 Remington Street Apt 425
Fort Collins, CO 80524

Date Mailed: 02/26/2026

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to the above-identified provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- A provisional application cover sheet was not provided for this application. The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying the application as a provisional application and providing the name of each inventor, either city and state, or city and foreign country, of the residence of each inventor, and the title of the invention.

  If the applicant wishes to submit an ADS to satisfy the cover sheet requirement, the ADS must be properly signed and properly marked up in compliance with 37 CFR 1.76(c), i.e. identifying the information that is being changed, with underlining for insertions, and strike-through or brackets for text removed.
- The statutory basic filing fee has not been received.
- Surcharge as set forth in 37 CFR 1.16(g) has not been received.

  The surcharge is due for late submission of filing fee or cover sheet.

SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- $**325** basic filing fee.
- $**65** surcharge.
- $(**0**) previous unapplied payment amount.
- $**390** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For

more information and a suggested format, see Form PTO/SB/92, MPEP 512, and www.uspto.gov/incomplete-application.

Replies should be addressed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

REPLY VIA ELECTRONIC FILING

A reply may be submitted via electronic filing only by registered users of the appropriate USPTO patent electronic filing system and must include:

- A separate document having the document description, "**Applicant response to Pre-Exam Formalities Notice**"; and
- the item(s) for which a reply period is set forth above.

For more information about electronic filing, contact the Patent Electronic Business Center (EBC) 1-866-217-9197 or visit https://www.uspto.gov/ebc

NOTE: If the reply is not being submitted electronically, a copy of this notice must be included.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/slee/

Welcome to the U.S. Patent and Trademark Office (USPTO)

*Congratulations on taking the first step to protect your invention. You can now advertise or mark any products covered by this application for your invention as **"Patent Pending"** for as long as this application is pending!*

We are here to help you on the rest of your journey.

New to the process? The Patent Basics page will take you from "What is a patent?" to assistance with the application process.

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Learn how to identify which of your creative ideas might be intellectual property assets and how to protect them by using the USPTO Intellectual Property (IP) Identifier.

- **ipidentifier.uspto.gov**

We also provide Pro Se Assistance, support specifically geared for **independent inventors without an attorney or agent.** Follow the link below, call us at 866-767-3848, or email **proseassistancecenter@uspto.gov**.

- **www.uspto.gov/ProSePatents**

Need help near you? We have regional offices across the country, as well as Patent and Trademark Resource Centers in libraries in nearly every state, with specialists standing by to help you at no cost.

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TO: E@headyconnection.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 221639

Thu Feb 26 05:12:05 EST 2026

Dear Patent Center Customer:

Correspondence Address:

Eric Haywood
149 Remington Street Apt 425
Fort Collins,COLORADO,80524
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 221639, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

The official date of notification of the outgoing correspondence will be indicated on the form (e.g., PTOL-90) accompanying the correspondence.

Disclaimer:
The list of documents shown below are provided as a courtesy and is not part of the official file wrapper. The content of the images shown in the Image File Wrapper is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
63965450	M327	02/26/2026	
63965450	APP.FILE.REC	02/26/2026	
63965450	PD.AUTH.NTC	02/26/2026	
63965450	SREXR.NTC	02/26/2026	
63965450	NTC.MISS.PRT	02/26/2026	
63965450	WELCOME.LET	02/26/2026	

To view your correspondence online, please sign in to Patent Center and then select Workbench/View correspondence. To update your email address(es), select Manage/Manage customer numbers.

If you have any questions, please contact the Patent Electronic Business Center (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

Please do not reply to this email as it was sent from an unmonitored mailbox.

Sincerely,

The Patent Center Team

UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/965,450	**01/22/2026 06:55:48 AM Z ET**	**-**

Title of Invention

System Efficiency Optimization via Spectral Analysis of Operational Frequencies and Constructive Interference Tuning

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	7115	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74086043	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74086043.pdf	5	Application Data Sheet	111 KB
PPA-031_HeadyResonance_SMALL_ENTITY_FINAL_v6-APP.TEXT.docx	11	Application body structured text document	535 KB

Warning: The specification paragraph numbering is not consecutive. Please review and revise if necessary. Figures have been detected in the specification. Please review and revise if necessary.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)

generatedADS74086043.pdf	483A27EFCB75CD16F3A851ABC3033BB36D246E0C5A0215E67805090BC40ADD334B3CA100A6DBD48A792B7E98D859C14E11A5284D935EF52B687FEDD908C8DA6A
PPA-031_HeadyResonance_SMALL_ENTITY_FINAL_v6-APP.TEXT.docx	3F9D7C929B8C5AFD6007B351DF0F501A66AEDCA53A25F0112A4A509DAADFF2BBCA58205F3DFE42C5B6F0AC8F3A36D0DC694691CB93A9827806D4177FD186BFD6

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

Residence Information
US Residency
Fort Collins, CO
UNITED STATES

Mailing Address
149 Remington Street Apt 425
Fort Collins, CO 80524
UNITED STATES

Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**System Efficiency Optimization via Spectral Analysis of Operational Frequencies and Constructive Interference Tuning**
Attorney docket number	**---**
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
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Applicant Information

of applicants: 1

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

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Applicant	Organization
Applicant type	Assignee

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

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Organization

Signature

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Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	01/22/2026

PROVISIONAL PATENT APPLICATION

System Efficiency Optimization via Spectral Analysis of Operational Frequencies and Constructive Interference Tuning

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Entity Status	Small Entity
U.S. Provisional Application No.	TBD
Government Interest	None
Attorney Docket	PPA-031
Date	January 22, 2026

System Efficiency Optimization via Spectral Analysis of Operational Frequencies and Constructive Interference Tuning

ABSTRACT

[0001] A system optimization engine treats digital and physical process telemetry as waveform data and uses spectral analysis to identify interference patterns among concurrent operations. A spectral monitor converts time-stamped logs, counters, and sensor streams into continuous signals and analyzes frequency-domain features to detect destructive interference associated with latency, heat, jitter, vibration, or throughput collapse. A phase-shift controller applies bounded micro-delays, pacing adjustments, and phase alignment to concurrent processes so their operational peaks align into a constructive interference state, improving throughput and reducing energy consumption without changing the underlying task logic. A dissonance alarm predicts impending gridlock or hardware failure by recognizing unstable or escalating destructive frequency signatures and can trigger mitigation actions, throttles, maintenance tickets, or safe-mode transitions.

CROSS-REFERENCE TO RELATED APPLICATIONS

[0002] This provisional application claims no priority to other applications at the time of filing, unless subsequently amended.

TECHNICAL FIELD

[0003] The present disclosure relates to high-performance computing optimization, cyber-physical system control, spectral analysis, timing control, and waveform engineering for improving efficiency and reliability of multi-process systems.

BACKGROUND

[0004] Modern computing and cyber-physical platforms execute large numbers of concurrent operations, including CPU scheduling, network I/O, storage access, interrupt servicing, control loops, and physical actuation.

[0005] Although each operation may be individually correct, aggregate behavior can become inefficient when process timings interact, producing emergent oscillations, queue resonance, thermal cycling, vibration amplification, or periodic contention.

[0006] Conventional optimizers treat telemetry as discrete events and typically tune parameters using heuristics, static thresholds, or coarse feedback controllers, which can miss frequency-domain interactions and phase relationships.

[0007] There is a need for an optimization approach that detects destructive timing interference across heterogeneous subsystems and applies bounded phase adjustments to create a stable, resonant operational regime with higher throughput and lower energy consumption.

SUMMARY OF THE INVENTION

[0008] In embodiments, telemetry from digital and physical subsystems is normalized into one or more time-aligned signals representing operational activity as a waveform or set of waveforms.

[0009] A spectral monitor computes frequency-domain representations, including fast Fourier transforms (FFT), short-time Fourier transforms (STFT), wavelet transforms, cross-spectral density, coherence, and phase difference metrics across concurrently operating components.

[0010] Destructive interference is identified where spectral energy, phase mismatch, or coherence patterns correlate with adverse outcomes such as increased latency, oscillatory queue depth, elevated temperatures, vibration, or increased power draw.

[0011] A phase-shift controller computes bounded timing offsets and pacing adjustments that alter the phase of selected processes, threads, network queues, or physical actuators to

improve coherence and align peaks into a constructive interference regime while respecting constraints such as deadlines, safety limits, and service-level objectives.

[0012] A dissonance alarm detects pre-failure signatures, including increasing phase jitter, spectral drift, coherence collapse, or harmonic growth, and triggers preemptive mitigation or maintenance actions.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 illustrates an example system architecture in which telemetry sources provide digital and physical signals to a spectral optimization engine and phase-shift controller.

[0014] FIG. 2 illustrates an example closed-loop control flow for spectral monitoring, interference classification, phase-shift actuation, and dissonance alarming.

FIG. 1 — Example Spectral Optimization Engine Architecture

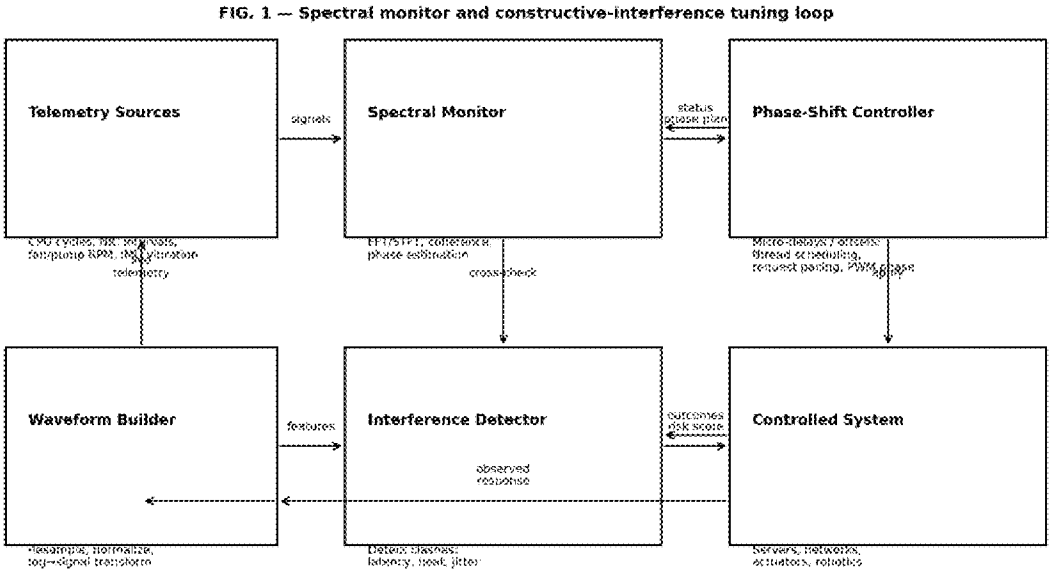
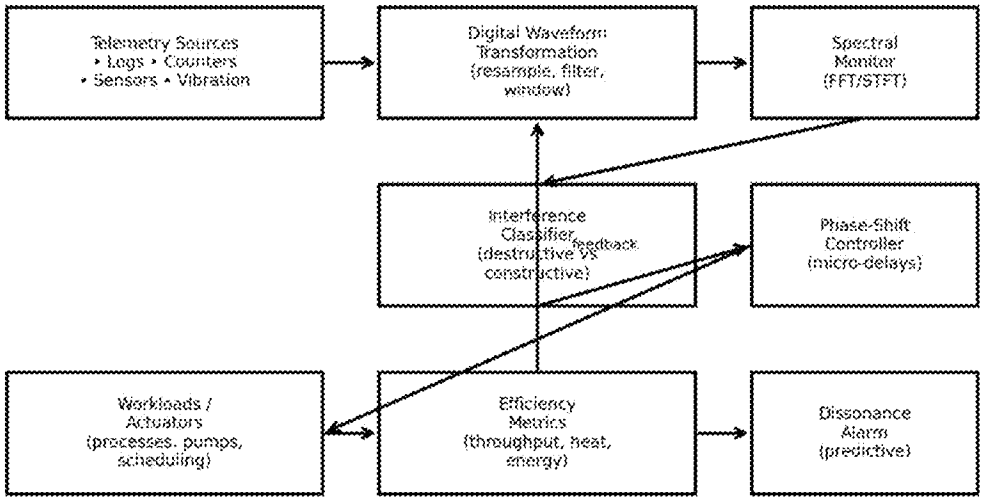


FIG. 2 — Closed-Loop Spectral Phase-Shift Control



DETAILED DESCRIPTION OF EMBODIMENTS

[0015] The following description is provided to enable a person skilled in the art to make and use the disclosed embodiments and sets forth various examples, variants, and implementation details.

[0016] The disclosed techniques may be implemented in software, firmware, hardware, or any combination thereof, including as an agent running on an operating system, a controller in an embedded platform, or a distributed service coordinating multiple nodes.

[0017] For clarity of explanation, the term "process" may refer to an operating-system process, a thread, a container workload, a network flow, a control loop, a scheduled job, or a physical actuator command stream, and the term "telemetry" may refer to any time-stamped measurement or event series.

Definitions

[0018] "Waveform transformation" refers to converting discrete telemetry (events, counters, logs, timestamps) into a time-domain signal suitable for spectral analysis, such as a uniformly sampled series, impulse train, or band-limited reconstruction.

[0019] "Spectral monitor" refers to a component that computes frequency-domain features from one or more waveforms and can include FFT, STFT, wavelets, cross-spectral methods, coherence estimation, and phase extraction.

[0020] "Destructive interference" refers to timing or frequency interactions among concurrent processes that correlate with degraded performance, instability, elevated power/temperature, vibration, or unreliable operation.

[0021] "Constructive interference" refers to timing or frequency alignment among concurrent processes that correlates with improved throughput, reduced energy use, reduced jitter, lower thermal stress, or reduced vibration.

[0022] "Phase-shift controller" refers to a controller that modifies timing of selected processes by applying bounded offsets, pacing, or scheduling adjustments to influence phase relationships without changing underlying task semantics.

Digital Waveform Transformation

[0023] In embodiments, telemetry includes one or more of: scheduler trace events, context switch timestamps, CPU cycle counters, cache-miss rates, network packet inter-arrival times, queue depths, I/O completion times, interrupt rates, fan RPM, pump RPM, vibration sensor streams, temperature sensor streams, and actuator command timings.

[0024] Telemetry that is event-based can be represented as a point process and transformed into an impulse train, where each event contributes an impulse at its timestamp with amplitude optionally weighted by event type, duration, or cost.

[0025] Telemetry that is counter-based can be converted into a derivative or rate signal by differencing successive samples and normalizing by elapsed time.

[0026] The waveform transformation may include resampling to a uniform sampling rate, time alignment across sources, normalization, windowing, smoothing kernels, anti-alias filtering, and segmentation into analysis windows.

[0027] Multiple telemetry streams can be combined into a multivariate signal, or separate waveforms can be maintained per component and later related through cross-spectral analysis.

Spectral Monitor

[0028] The spectral monitor computes frequency-domain representations of one or more waveforms, including magnitude spectra, phase spectra, spectrograms, and time-frequency decompositions.

[0029] In one embodiment, the spectral monitor computes an STFT over sliding windows and estimates per-window dominant frequencies, harmonic structure, and spectral entropy.

[0030] In another embodiment, a wavelet transform is used to detect transient oscillations and non-stationary components that may be missed by fixed-window Fourier methods.

[0031] For interactions between components, cross-spectral density and coherence can be computed to quantify coupling strength and phase alignment between two signals.

[0032] The spectral monitor may maintain baseline models for normal operation and compute deviations, including spectral drift, emergent harmonics, coherence loss, or increasing phase noise.

Interference Classification and Optimization Objective

[0033] Destructive interference can manifest as periodic contention (e.g., synchronized wake-ups causing cache thrash), queue resonance (e.g., bursty arrivals aligning with service cycles), thermal cycling (e.g., alternating workloads exciting heat inertia), or mechanical vibration (e.g., pump pulses aligning with structural modes).

[0034] Embodiments compute an interference score that aggregates one or more metrics, including coherence at problematic bands, phase mismatch, queue-depth oscillation amplitude, temperature oscillation amplitude, and latency variance.

[0035] Embodiments compute a constructive score that reflects desired alignment, such as improved coherence between workload pacing and cooling cycles, reduced jitter, reduced spectral energy in known resonance bands, or increased throughput per watt.

[0036] The optimizer selects phase adjustments that reduce the interference score and/or increase the constructive score subject to constraints such as deadlines, fairness, and safety limits.

[0037] The classifier may be rule-based, statistical, or machine-learning-based, and may incorporate domain knowledge such as known resonant frequencies of a chassis or known scheduler tick rates.

Phase-Shift Controller

[0038] The phase-shift controller applies bounded timing modifications to selected processes or actuators to influence phase relationships, including micro-delays, pacing changes, or schedule offsets.

[0039] In a computing system, phase shifting can be implemented using mechanisms such as sleep or busy-wait offsets, timer slack, task affinity, cgroup bandwidth limits, scheduler latency tuning, interrupt coalescing parameters, and network queue pacing.

[0040] In a cyber-physical system, phase shifting can be implemented by adjusting PWM phases, motor commutation timing, pump duty-cycle offsets, or control loop sample times within safe bounds.

[0041] The controller can operate as a closed-loop optimizer that periodically re-estimates spectral features, evaluates interference scores, and updates offsets, with hysteresis and rate limits to avoid instability.

[0042] Embodiments restrict phase adjustments to be semantics-preserving, for example by only inserting delays that do not violate explicit deadlines, by respecting real-time scheduling constraints, and by limiting phase modifications to slack intervals identified by the system.

Example Control Loop (Pseudocode)

[0043] An example procedure is provided below to illustrate one embodiment of a spectral optimization control loop.

[0044] Procedure Optimize():

```
 while system_running:
   T <- CollectTelemetry(window=W)
   X <- WaveformTransform(T, fs)
   S <- SpectralFeatures(X)
   I <- InterferenceScore(S, baselines)
   if I > threshold:
     Δ <- SolvePhaseAdjustments(S, constraints)
     ApplyPhaseShift(Δ)
   A <- DissonanceAlarm(S, trend_models)
   if A == TRIGGER:
     MitigateOrNotify()
   sleep(control_period)
```

Dissonance Alarm and Predictive Maintenance

[0045] The dissonance alarm identifies destructive spectral signatures that are predictive of impending failure or gridlock, including growing resonance peaks, increasing phase noise, coherence collapse, or spectral drift beyond a confidence band.

[0046] In embodiments, the alarm computes a trend over successive windows and triggers when a slope, variance, or likelihood exceeds a threshold, enabling early mitigation before user-visible downtime.

[0047] The alarm can trigger actions such as dynamic throttling, load shedding, migration of workloads, fan/pump phase re-timing, or creation of a maintenance ticket associated with a specific component or subsystem.

[0048] In a physical platform, the alarm can initiate a safe-mode that reduces actuation energy at resonant frequencies and prevents structural amplification, while also recording a forensic evidence bundle for later diagnosis.

Representative Use Cases

[0049] Data center and edge compute: aligning batch job start times, network polling cycles, and cooling equipment cycles to reduce periodic thermal peaks and improve throughput per watt.

[0050] HeadyCool cooling orchestration: aligning pump cycles, chiller duty cycles, and workload scheduling so heat generation and heat removal are phase-aligned, reducing thermal oscillation and mechanical wear.

[0051] HeadySpace microgravity and vibration control: detecting resonant vibrations from fans, pumps, reaction wheels, or robotic actuation and phase-shifting actuation commands to avoid structural resonance in low-damping environments.

[0052] Industrial automation: phase aligning conveyance, robotic motion, and vision processing to reduce jitter, prevent queue resonance, and increase line throughput without changing control logic.

Implementation Variations

[0053] Centralized controller: a single optimization engine ingests telemetry across components and applies phase shifts via a unified policy layer.

[0054] Distributed controller: per-node spectral monitors compute local features and share compact summaries to coordinate phase alignment across a cluster.

[0055] Hierarchical controller: local phase controllers manage fast dynamics while a global optimizer manages slower scheduling and cooling cycles.

[0056] Hybrid digital-physical controller: the same spectral framework is applied to both digital scheduling and physical actuation, enabling coordinated resonance tuning across cyber and physical domains.

Advantages

[0057] Improves throughput and reduces energy consumption by treating concurrency as a coupled waveform system and correcting phase interactions rather than tuning only static parameters.

[0058] Reduces thermal cycling and vibration stress by identifying resonant harmonics and suppressing destructive frequency patterns through bounded phase shifts.

[0059] Enables predictive maintenance using frequency-domain signatures that often appear before threshold-based alarms in time-domain metrics.

[0060] Provides a general framework applicable to heterogeneous systems, including servers, networks, storage, robotics, and cooling infrastructure.

NON-LIMITING EXAMPLE CLAIMS

[0061] A method of optimizing a system comprising: collecting telemetry from a plurality of concurrent operations; converting the telemetry into one or more time-domain signals; computing spectral features from the one or more time-domain signals; identifying a destructive interference pattern based on the spectral features; and applying a phase adjustment to at least one operation to reduce the destructive interference pattern.

[0062] The method of claim 1, wherein converting the telemetry into the one or more time-domain signals includes transforming event timestamps into an impulse train and resampling the impulse train to a uniform sampling rate.

[0063] The method of claim 1, wherein computing spectral features includes at least one of an FFT, an STFT, a wavelet transform, cross-spectral density, coherence estimation, or phase difference estimation.

[0064] The method of claim 1, wherein identifying the destructive interference pattern includes detecting a resonance band associated with increased latency variance, queue oscillation, thermal oscillation, vibration amplitude, or power draw.

[0065] The method of claim 1, wherein applying the phase adjustment comprises applying a bounded micro-delay to a process, thread, job, or network flow to align an operational peak with an operational peak of another process, thread, job, or network flow.

[0066] The method of claim 5, wherein the bounded micro-delay is selected subject to constraints that preserve task semantics, deadlines, or service-level objectives.

[0067] The method of claim 1, further comprising producing a dissonance alarm based on a trend in spectral features indicative of impending hardware failure or software gridlock.

[0068] The method of claim 7, wherein producing the dissonance alarm includes detecting increasing phase jitter, coherence collapse, spectral drift, or growth of harmonic energy over successive analysis windows.

[0069] A system comprising: a telemetry collector; a waveform transformation engine configured to convert telemetry into one or more signals; a spectral monitor configured to compute frequency-domain features from the one or more signals; and a phase-shift controller configured to apply timing adjustments that increase constructive interference among concurrent operations.

[0070] The system of claim 9, wherein the phase-shift controller modifies at least one of scheduler parameters, interrupt coalescing parameters, network queue pacing, PWM phase timing, motor commutation timing, or pump duty-cycle timing.

[0071] The system of claim 9, further comprising a predictive maintenance module configured to generate an alert when the spectral monitor identifies a destructive pattern associated with impending failure.

[0072] A non-transitory computer-readable medium storing instructions that, when executed, cause a processor to perform the method of any of claims 1–8.

CONCLUSION

[0073] While the disclosed embodiments have been described with reference to specific examples, the scope of the disclosure is not limited to those examples, and variations may be made without departing from the spirit of the disclosure.